

# Montrac RICAIP Maze Assembly Plan

## Manual

Description of operations needed to assemble the RICAIP maze.

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W1/W2 Montrac

V2

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## PLAN

|     |                                              |    |
|-----|----------------------------------------------|----|
| 1   | Place plastic inserts .....                  | 3  |
| 1.1 | Pick "START" insert .....                    | 3  |
| 1.2 | Place "START" insert .....                   | 3  |
| 1.3 | Pick "FINISH" insert .....                   | 4  |
| 1.4 | Place "FINISH" insert .....                  | 4  |
| 1.5 | Pick "LOGO" insert .....                     | 5  |
| 1.6 | Place "LOGO" insert .....                    | 5  |
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| 2.3 | Dispense ball bearing 2 .....                | 7  |
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| 4.4 | Place rivet 2 .....                          | 10 |
| 4.5 | Pick rivet 3 .....                           | 11 |
| 4.6 | Place rivet 3 .....                          | 11 |

|      |                       |    |
|------|-----------------------|----|
| 4.7  | Pick rivet 4 .....    | 12 |
| 4.8  | Place rivet 4 .....   | 12 |
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| 4.11 | Push in rivet 2 ..... | 13 |
| 4.12 | Push in rivet 3 ..... | 14 |
| 4.13 | Push in rivet 4 ..... | 14 |

## STEP 1 - Place plastic inserts

Allowed robots: R1

### SUBSTEP 1.1 - Pick "START" insert

Operation - ID 21 - Pick Vertical

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 70,000  |

### SUBSTEP 1.2 - Place "START" insert

Operation - ID 31 - Place Vertical

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 10,000  |

**SUBSTEP 1.3 - Pick “FINISH” insert****Operation - ID 21 - Pick Vertical**

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 70,000  |

**SUBSTEP 1.4 - Place “FINISH” insert****Operation - ID 31 - Place Vertical**

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 10,000  |

**SUBSTEP 1.5 - Pick “LOGO” insert****Operation - ID 21 - Pick Vertical**

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 70,000  |

**SUBSTEP 1.6 - Place “LOGO” insert****Operation - ID 31 - Place Vertical**

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 10,000  |

## STEP 2 - Insert ball bearings

Allowed robots: R3

### SUBSTEP 2.1 - Pick ball bearing dispenser from stand

Operation - ID 21 - Pick Vertical

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 80,000  |

### SUBSTEP 2.2 - Dispense ball bearing 1

Operation - ID 41 - Approach Vertical

| Table of operation parameters |         |         |
|-------------------------------|---------|---------|
| Index                         | Name    | Value   |
| 0                             | X       | Unknown |
| 1                             | Y       | Unknown |
| 2                             | Z       | Unknown |
| 3                             | A       | Unknown |
| 4                             | Speed   | 100     |
| 5                             | Base ID | Unknown |
| 6                             | Return  | 0       |
| 7                             | Offset  | 200,000 |

**SUBSTEP 2.3 - Dispense ball bearing 2****Operation - ID 41 - Approach Vertical**

| Table of operation parameters |         |         |
|-------------------------------|---------|---------|
| Index                         | Name    | Value   |
| 0                             | X       | Unknown |
| 1                             | Y       | Unknown |
| 2                             | Z       | Unknown |
| 3                             | A       | Unknown |
| 4                             | Speed   | 100     |
| 5                             | Base ID | Unknown |
| 6                             | Return  | 0       |
| 7                             | Offset  | 200,000 |

**SUBSTEP 2.4 - Return ball bearing dispenser to stand****Operation - ID 31 - Place Vertical**

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 80,000  |

## STEP 3 - Place plexiglass cover

Allowed robots: R2

### SUBSTEP 3.1 - Pick plexiglass

Operation - ID 21 - Pick Vertical

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | N/A     |

### SUBSTEP 3.2 - Place plexiglass

Operation - ID 31 - Place Vertical

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | N/A     |



## STEP 4 - Insert rivets

Allowed robots: R1

### SUBSTEP 4.1 - Pick rivet 1

Operation - ID 21 - Pick Vertical

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 10,000  |

### SUBSTEP 4.2 - Place rivet 1

Operation - ID 31 - Place Vertical

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 10,000  |

**SUBSTEP 4.3 - Pick rivet 2****Operation - ID 21 - Pick Vertical**

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 10,000  |

**SUBSTEP 4.4 - Place rivet 2****Operation - ID 31 - Place Vertical**

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 10,000  |

**SUBSTEP 4.5 - Pick rivet 3****Operation - ID 21 - Pick Vertical**

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 10,000  |

**SUBSTEP 4.6 - Place rivet 3****Operation - ID 31 - Place Vertical**

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 10,000  |

**SUBSTEP 4.7 - Pick rivet 4****Operation - ID 21 - Pick Vertical**

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 10,000  |

**SUBSTEP 4.8 - Place rivet 4****Operation - ID 31 - Place Vertical**

| Table of operation parameters |                             |         |
|-------------------------------|-----------------------------|---------|
| Index                         | Name                        | Value   |
| 0                             | X                           | Unknown |
| 1                             | Y                           | Unknown |
| 2                             | Z                           | Unknown |
| 3                             | A                           | Unknown |
| 4                             | Speed                       | 100     |
| 5                             | Base ID                     | Unknown |
| 6                             | Return                      | 0       |
| 7                             | Offset                      | 200,000 |
| 8                             | Gripper<br>Open<br>Position | 10,000  |

**SUBSTEP 4.9 - Close gripper****Operation - ID 51 - Run Tool Command**

| Table of operation parameters |                 |       |
|-------------------------------|-----------------|-------|
| Index                         | Name            | Value |
| 0                             | Gripper Command | 21    |

**SUBSTEP 4.10 - Push in rivet 1****Operation - ID 41 - Approach Vertical**

| Table of operation parameters |         |         |
|-------------------------------|---------|---------|
| Index                         | Name    | Value   |
| 0                             | X       | Unknown |
| 1                             | Y       | Unknown |
| 2                             | Z       | Unknown |
| 3                             | A       | Unknown |
| 4                             | Speed   | 100     |
| 5                             | Base ID | Unknown |
| 6                             | Return  | 0       |
| 7                             | Offset  | 200,000 |

**SUBSTEP 4.11 - Push in rivet 2****Operation - ID 41 - Approach Vertical**

| Table of operation parameters |         |         |
|-------------------------------|---------|---------|
| Index                         | Name    | Value   |
| 0                             | X       | Unknown |
| 1                             | Y       | Unknown |
| 2                             | Z       | Unknown |
| 3                             | A       | Unknown |
| 4                             | Speed   | 100     |
| 5                             | Base ID | Unknown |
| 6                             | Return  | 0       |
| 7                             | Offset  | 200,000 |

**SUBSTEP 4.12 - Push in rivet 3****Operation - ID 41 - Approach Vertical**

| Table of operation parameters |         |         |
|-------------------------------|---------|---------|
| Index                         | Name    | Value   |
| 0                             | X       | Unknown |
| 1                             | Y       | Unknown |
| 2                             | Z       | Unknown |
| 3                             | A       | Unknown |
| 4                             | Speed   | 100     |
| 5                             | Base ID | Unknown |
| 6                             | Return  | 0       |
| 7                             | Offset  | 200,000 |

**SUBSTEP 4.13 - Push in rivet 4****Operation - ID 41 - Approach Vertical**

| Table of operation parameters |         |         |
|-------------------------------|---------|---------|
| Index                         | Name    | Value   |
| 0                             | X       | Unknown |
| 1                             | Y       | Unknown |
| 2                             | Z       | Unknown |
| 3                             | A       | Unknown |
| 4                             | Speed   | 100     |
| 5                             | Base ID | Unknown |
| 6                             | Return  | 0       |
| 7                             | Offset  | 200,000 |